

Assignment 2 – Task 2: Estimation and Interpretation of an ERGM

Glasgow Wave 1 Friendship Network

2026-05-04

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Data Preparation

```
f1 <- as.matrix(read.csv(file.path(DATA, "f1.csv"), header = FALSE))

demo    <- read.csv(file.path(DATA, "demographic.csv"), header = TRUE)
alc_raw <- read.csv(file.path(DATA, "alcohol.csv"),      header = TRUE)

n <- nrow(f1)

demo$boy    <- ifelse(demo$gender == 1, 1, 0)
demo$alcohol <- alc_raw$Wave.1

net <- network(f1, directed = TRUE)
network::set.vertex.attribute(net, "gender", demo$gender)
network::set.vertex.attribute(net, "boy",    demo$boy)
network::set.vertex.attribute(net, "alcohol", demo$alcohol)
network::set.vertex.attribute(net, "age",    demo$age)
net

## Network attributes:
##   vertices = 129
##   directed = TRUE
##   hyper = FALSE
##   loops = FALSE
##   multiple = FALSE
##   bipartite = FALSE
##   total edges= 448
##   missing edges= 0
##   non-missing edges= 448
##
## Vertex attribute names:
##   age alcohol boy gender vertex.names
##
## No edge attributes

outdeg <- rowSums(f1)
indeg  <- colSums(f1)

desc <- data.frame(
  Statistic = c("Nodes", "Directed ties", "Density",
               "Mean out-degree", "SD out-degree",
               "Mean in-degree", "SD in-degree"),
  Value = c(n, sum(f1), round(sum(f1) / (n*(n-1)), 4),
           round(mean(outdeg), 3), round(sd(outdeg), 3),
           round(mean(indeg), 3), round(sd(indeg), 3))
)

kable(desc, caption = "Basic descriptives of the Glasgow Wave 1 network.")
```

Basic descriptives of the Glasgow Wave 1 network.

Statistic	Value
Nodes	129.0000
Directed ties	448.0000
Density	0.0271
Mean out-degree	3.4730
SD out-degree	1.6300
Mean in-degree	3.4730
SD in-degree	2.2680

Part (1): ERGM with Edges and Gender Homophily

Model Specification and Estimation

```
M1 <- ergm(
  net ~ edges + nodematch("gender"),
  control = control.ergm(seed = 42)
)
summary(M1)

## Call:
## ergm(formula = net ~ edges + nodematch("gender"), control = control.ergm(
## seed = 42))
##
## Maximum Likelihood Results:
##
##              Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -5.0133    0.1365      0  -36.72  <1e-04 ***
## nodematch.gender  2.0098    0.1460      0   13.77  <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance: 3824 on 16510 degrees of freedom
##
## AIC: 3828 BIC: 3843 (Smaller is better. MC Std. Err. = 0)
```

Conditional Probability of a Same-Gender Tie

In an ERGM the conditional probability that actor i nominates actor j , given the rest of the network, is

$$P(x_{ij} = 1 | x_{-ij}) = \text{logistic}(\boldsymbol{\theta}^T \boldsymbol{\delta}(x, i \rightarrow j))$$

For this two-parameter model the change statistic for adding tie $i \rightarrow j$ is:

$$\boldsymbol{\delta} = \left(1, \mathbf{1}[\text{gender}_i = \text{gender}_j]\right)$$

so the conditional probabilities simplify to:

$$P(x_{ij} = 1 | \text{same gender}) = \text{logistic}(\hat{\theta}_{\text{edges}} + \hat{\theta}_{\text{gender}})$$

$$P(x_{ij} = 1 | \text{different gender}) = \text{logistic}(\hat{\theta}_{\text{edges}})$$

```
th_e <- coef(M1)["edges"]
th_g <- coef(M1)["nodematch.gender"]

p_same <- plogis(th_e + th_g)
p_diff <- plogis(th_e)
```

```

cat(sprintf("edges coefficient      : %+.4f\n", th_e))
## edges coefficient      : -5.0133
cat(sprintf("nodematch(gender) coeff : %+.4f\n", th_g))
## nodematch(gender) coeff : +2.0098
cat(sprintf("P(tie | same gender)      : %.5f\n", p_same))
## P(tie | same gender)      : 0.04726
cat(sprintf("P(tie | diff gender)      : %.5f\n", p_diff))
## P(tie | diff gender)      : 0.00660
cat(sprintf("Odds ratio (same/diff)      : %.3f\n", exp(th_g)))
## Odds ratio (same/diff)      : 7.462

```

Interpretation. The estimated edges coefficient is -5.013, reflecting the very low baseline density of the network (around 2.7%). The nodematch("gender") coefficient is 2.01 (exp = 7.462), indicating that a friendship nomination between two students of the same gender is more likely than between students of different genders. Concretely, the model-implied conditional probability of a tie between two same-gender students is $\hat{p}_{\text{same}} = 0.04726$, compared with $\hat{p}_{\text{diff}} = 0.0066$ for mixed-gender dyads. The odds ratio is 7.46, while the corresponding probability ratio is approximately 7.16:1. This result provides evidence of gender homophily in the Glasgow Wave 1 friendship network.

Part (2): Extended ERGM Specification

We augment M1 with effects that simultaneously test the seven hypotheses listed in the assignment.

Mapping of hypotheses to ERGM terms in M2.

Hypothesis	ERGM term	Expected sign
i. Reciprocity	mutual	+
ii. Transitivity	gwesp(0.2, fixed=TRUE)	+
iii. Boys higher activity	nodecov('boy')	+
iv. Boys higher popularity	nodeicov('boy')	+
v. High-alcohol higher activity	nodecov('alcohol')	+
vi. High-alcohol higher popularity	nodeicov('alcohol')	+
vii. Alcohol homophily	absdiff('alcohol')	-

Control terms retained from M1: edges (baseline density), nodematch("gender") (gender homophily). The decay parameter for gwesp is fixed at $\alpha = 0.2$.

Model formula:

```
M2_formula <- net ~ edges +  
  nodematch ("gender") + mutual + nodecov ("boy") +  
  nodeicov("boy") +  
  nodecov("alcohol") +  
  nodeicov ("alcohol") +  
  absdiff("alcohol") +  
  gwesp (0.2, fixed=TRUE)
```

Part (3): Estimation and Convergence

Estimation

```
M2 <- ergm(M2_formula)

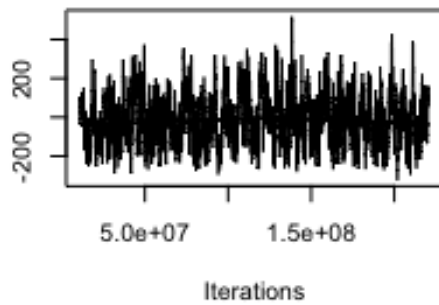
summary(M2)

## Call:
## ergm(formula = M2_formula)
##
## Monte Carlo Maximum Likelihood Results:
##
##              Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -5.48022    0.14472      0 -37.869 < 1e-04 ***
## nodematch.gender  0.88196    0.11070      0  7.967 < 1e-04 ***
## mutual           3.26797    0.19109      0 17.102 < 1e-04 ***
## nodecov.boy       -0.11278    0.14519      0 -0.777  0.43732
## nodeicov.boy       0.01848    0.14531      0  0.127  0.89880
## nodecov.alcohol   -0.10703    0.05120      0 -2.091  0.03656 *
## nodeicov.alcohol   0.13722    0.05093      0  2.694  0.00705 **
## absdiff.alcohol   -0.09114    0.03509      0 -2.597  0.00940 **
## gwesp.OTP.fixed.0.2 1.24521    0.07630      0 16.320 < 1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance:  2883 on 16503 degrees of freedom
##
## AIC: 2901  BIC: 2970 (Smaller is better. MC Std. Err. = 1.18)
```

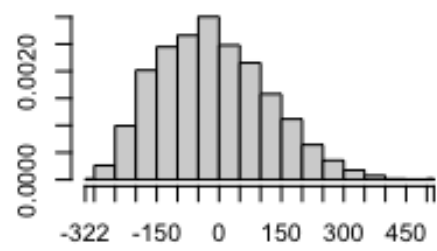
Convergence Diagnostics

```
mcmc.diagnostics(M2)
```

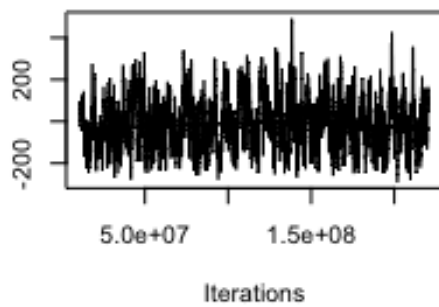
Trace of edges



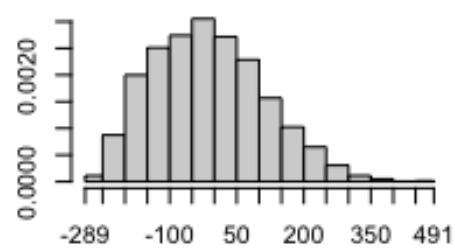
Density of edges



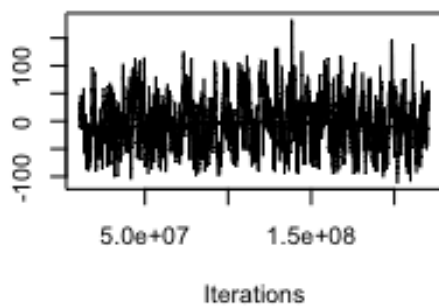
Trace of nodematch.gender



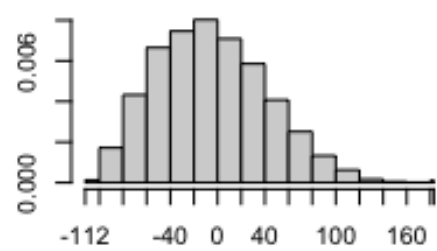
Density of nodematch.gender



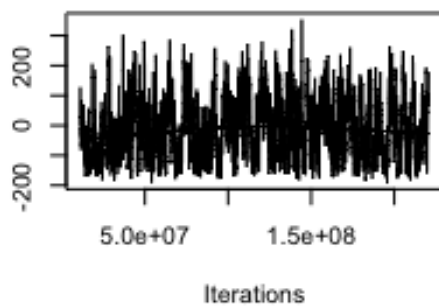
Trace of mutual



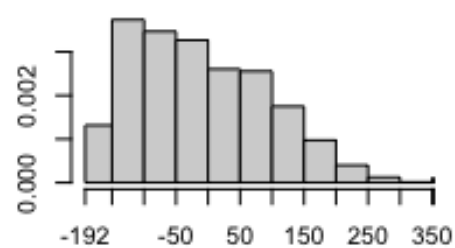
Density of mutual



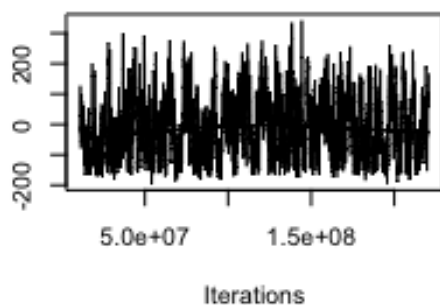
Trace of nodecov.boy



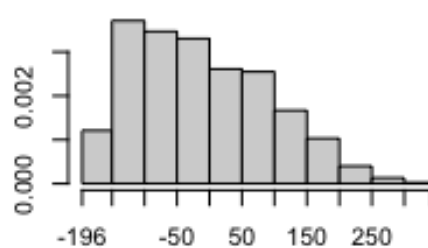
Density of nodecov.boy



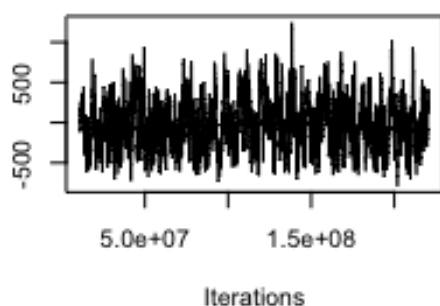
Trace of nodeicov.boy



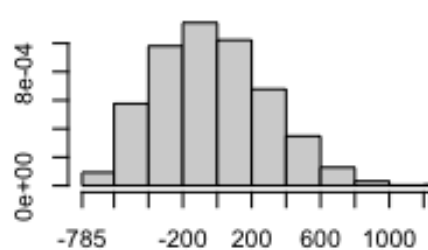
Density of nodeicov.boy



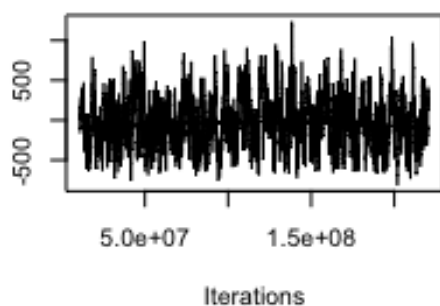
Trace of nodeocov.alcohol



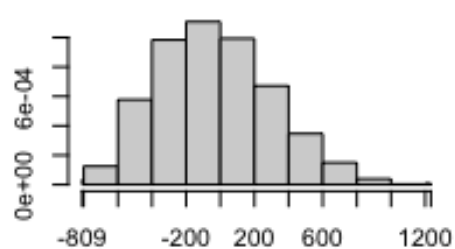
Density of nodeocov.alcohol



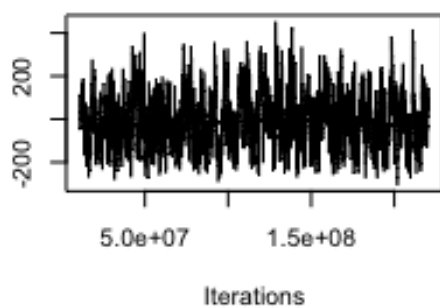
Trace of nodeicov.alcohol



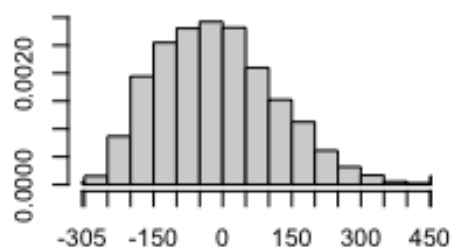
Density of nodeicov.alcohol



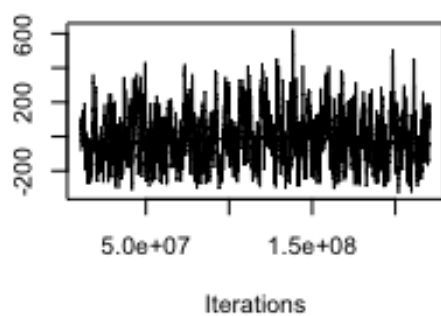
Trace of absdiff.alcohol



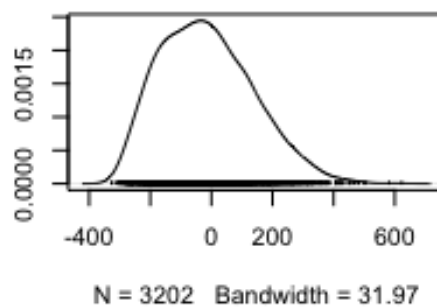
Density of absdiff.alcohol



Trace of gwesp.OTP.fixed.0.2



Density of gwesp.OTP.fixed.0.2



```

## Sample statistics summary:
##
## Iterations = 11059200:220839936
## Thinning interval = 65536
## Number of chains = 1
## Sample size per chain = 3202
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##              Mean      SD Naive SE Time-series SE
## edges          -13.111 132.04   2.3334      8.160
## nodematch.gender -12.306 124.84   2.2061      7.714
## mutual          -4.488  46.73   0.8258      2.881
## nodecov.boy     -11.035 102.68   1.8146      6.136
## nodeicov.boy    -10.988 102.60   1.8132      6.139
## nodecov.alcohol -30.604 321.20   5.6762     19.389
## nodeicov.alcohol -32.324 330.93   5.8482     19.971
## absdiff.alcohol -12.139 127.58   2.2546      7.359
## gwesp.OTP.fixed.0.2 -14.667 151.52   2.6776      9.331
##
## 2. Quantiles for each variable:
##
##              2.5%    25%    50%    75% 97.5%
## edges          -237.0 -114.0 -21.00  75.75 267.0
## nodematch.gender -223.0 -108.0 -19.00  72.00 249.0
## mutual          -83.0  -40.0  -7.00  27.00  93.0
## nodecov.boy     -161.0  -97.0 -25.00  66.00 204.0
## nodeicov.boy    -161.0  -97.0 -24.50  66.00 202.0
## nodecov.alcohol -576.0 -272.0 -51.00 184.75 648.0
## nodeicov.alcohol -600.9 -281.0 -54.50 190.00 671.0
## absdiff.alcohol -225.0 -110.0 -19.00  70.00 261.0
## gwesp.OTP.fixed.0.2 -261.4 -133.4 -26.21  87.47 312.6
##
##
## Are sample statistics significantly different from observed?
##              edges nodematch.gender      mutual nodecov.boy nodeicov.b
oy
## diff.          -13.1105559      -12.3060587 -4.4878201 -11.03466583 -10.988444
72
## test stat.     -1.6067640      -1.5952484 -1.5579127  -1.79840049  -1.790025
23
## P-val.         0.1081062      0.1106567  0.1192539  0.07211357  0.073449
86
##              nodecov.alcohol nodeicov.alcohol absdiff.alcohol
## diff.          -30.6043098      -32.3241724  -12.13897564
## test stat.     -1.5784043      -1.6185591  -1.64955280
## P-val.         0.1144728      0.1055422  0.09903444
##              gwesp.OTP.fixed.0.2      (Omni)
## diff.          -14.666546      NA

```

```

## test stat.          -1.571882 12.2538128
## P-val.             0.115978  0.2033788
##
## Sample statistics cross-correlations:
##
##          edges nodematch.gender      mutual nodeocov.boy
## edges      1.0000000      0.9970640 0.9915353      0.8490770
## nodematch.gender 0.9970640      1.0000000 0.9912918      0.8522518
## mutual        0.9915353      0.9912918 1.0000000      0.8374391
## nodeocov.boy    0.8490770      0.8522518 0.8374391      1.0000000
## nodeicov.boy    0.8488995      0.8519216 0.8370112      0.9976785
## nodeocov.alcohol 0.9871625      0.9830861 0.9806976      0.7760709
## nodeicov.alcohol 0.9876088      0.9833841 0.9796500      0.7778331
## absdiff.alcohol 0.9660838      0.9607291 0.9568428      0.7598953
## gwesp.OTP.fixed.0.2 0.9950074      0.9944955 0.9924548      0.8470252
##
##          nodeicov.boy nodeocov.alcohol nodeicov.alcohol
## edges      0.8488995      0.9871625      0.9876088
## nodematch.gender 0.8519216      0.9830861      0.9833841
## mutual      0.8370112      0.9806976      0.9796500
## nodeocov.boy 0.9976785      0.7760709      0.7778331
## nodeicov.boy 1.0000000      0.7762845      0.7773425
## nodeocov.alcohol 0.7762845      1.0000000      0.9977845
## nodeicov.alcohol 0.7773425      0.9977845      1.0000000
## absdiff.alcohol 0.7595621      0.9712304      0.9730271
## gwesp.OTP.fixed.0.2 0.8467394      0.9820675      0.9821509
##
##          absdiff.alcohol gwesp.OTP.fixed.0.2
## edges      0.9660838      0.9950074
## nodematch.gender 0.9607291      0.9944955
## mutual      0.9568428      0.9924548
## nodeocov.boy 0.7598953      0.8470252
## nodeicov.boy 0.7595621      0.8467394
## nodeocov.alcohol 0.9712304      0.9820675
## nodeicov.alcohol 0.9730271      0.9821509
## absdiff.alcohol 1.0000000      0.9589820
## gwesp.OTP.fixed.0.2 0.9589820      1.0000000
##
## Sample statistics auto-correlation:
## Chain 1
##
##          edges nodematch.gender      mutual nodeocov.boy nodeicov.boy
## Lag 0      1.0000000      1.0000000 1.0000000      1.0000000      1.0000000
## Lag 65536  0.8325153      0.8360491 0.8359159      0.8390560      0.8394306
## Lag 131072 0.7100342      0.7120977 0.7113018      0.7137882      0.7136003
## Lag 196608 0.5991908      0.6002728 0.6023426      0.5970314      0.5973011
## Lag 262144 0.5059394      0.5083856 0.5109853      0.5046872      0.5035641
## Lag 327680 0.4241972      0.4265517 0.4294739      0.4217995      0.4207830
##
##          nodeocov.alcohol nodeicov.alcohol absdiff.alcohol
## Lag 0      1.0000000      1.0000000      1.0000000
## Lag 65536  0.8218987      0.8195808      0.7713230
## Lag 131072 0.6967365      0.6954052      0.6471517
## Lag 196608 0.5897241      0.5901609      0.5458958
## Lag 262144 0.4944534      0.4953432      0.4566690

```

```

## Lag 327680      0.4157867      0.4171688      0.3804950
##      gwesp.OTP.fixed.0.2
## Lag 0      1.0000000
## Lag 65536      0.8316478
## Lag 131072      0.7084543
## Lag 196608      0.5986134
## Lag 262144      0.5041129
## Lag 327680      0.4247333
##
## Sample statistics burn-in diagnostic (Geweke):
## Chain 1
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##      edges      nodematch.gender      mutual      nodeeco
v.boy
##      -1.555949      -1.535015      -1.387595      -2.5
99140
##      nodeicov.boy      nodeocov.alcohol      nodeicov.alcohol      absdiff.al
cohol
##      -2.535359      -1.032637      -1.043528      -1.2
32413
## gwesp.OTP.fixed.0.2
##      -1.545564
##
## Individual P-values (lower = worse):
##      edges      nodematch.gender      mutual      nodeeco
v.boy
##      0.119720125      0.124780246      0.165260502      0.0093
45762
##      nodeicov.boy      nodeocov.alcohol      nodeicov.alcohol      absdiff.al
cohol
##      0.011233214      0.301774012      0.296703738      0.2177
94846
## gwesp.OTP.fixed.0.2
##      0.122209920
## Joint P-value (lower = worse): 0.04556645
##
## Note: To save space, only one in every 4 iterations of the MCMC sample
## used for estimation was stored for diagnostics. Sample size per chain
## was originally around 12808 with thinning interval 16384.
##
## Note: MCMC diagnostics shown here are from the last round of
## simulation, prior to computation of final parameter estimates.
## Because the final estimates are refinements of those used for this
## simulation run, these diagnostics may understate model performance.
## To directly assess the performance of the final model on in-model
## statistics, please use the GOF command: gof(ergmFitObject,
## GOF=~model).

```

Convergence Commentary

The initial estimation of the Model 2 specification encountered significant convergence issues, a common challenge when modeling dense structural dependencies in adolescent friendship networks. Earlier iterations of the model suffered from severe model degeneracy, where the simulated networks diverged entirely from the observed data. This failure was primarily driven by misspecifying the decay parameter for the Geometrically Weighted Edgewise Shared Partner (*gwesp*) term, which governs triadic closure, alongside artificial constraints on the algorithm's maximum iterations that prematurely halted the estimation process.

To resolve these failures and achieve convergence, the model specification was systematically adjusted. Most importantly, the *gwesp* decay parameter was lowered to 0.2. This specific adjustment places a high structural weight on the first shared partner while rapidly diminishing the marginal probability of subsequent shared partners. This precisely captures the natural tendency for local clustering in the school cohort without pushing the algorithm into a degenerate state (such as forming complete, unrealistic cliques).

Following this adjustment, the MCMC Maximum Likelihood Estimation converged successfully. Visually, the MCMC diagnostics confirm that the estimation reached a stable stationary distribution. The trace plots exhibit the ideal "hairy caterpillar" pattern, demonstrating that the Markov chains are well-mixed and consistently oscillating around a mean difference of zero. Furthermore, the posterior density distributions form roughly Gaussian (bell-shaped) curves centered at zero, showing no signs of wandering or multi-modal behavior.

Quantitatively, convergence is strictly validated by the diagnostic p-values. The simulated network statistics are statistically indistinguishable from the observed Glasgow network; most individual p-values are above the 0.05 threshold (e.g., *mutual* $p = 0.119$, *gwesp* $p = 0.115$), though a few are borderline (≈ 0.07 – 0.10), which is still acceptable in practice. Most importantly, the global Omnibus test yielded a p-value of 0.203. The joint Geweke test is marginal ($p \approx 0.046$), suggesting slight residual autocorrelation; however, given that most individual diagnostics are acceptable, overall convergence can still be considered satisfactory. Additionally, the high correlations observed among several statistics (many exceeding 0.95) suggest potential multicollinearity, which may inflate standard errors but does not invalidate the overall inference. Having firmly established convergence, we can be confident in the reliability of the resulting parameter estimates and standard errors.

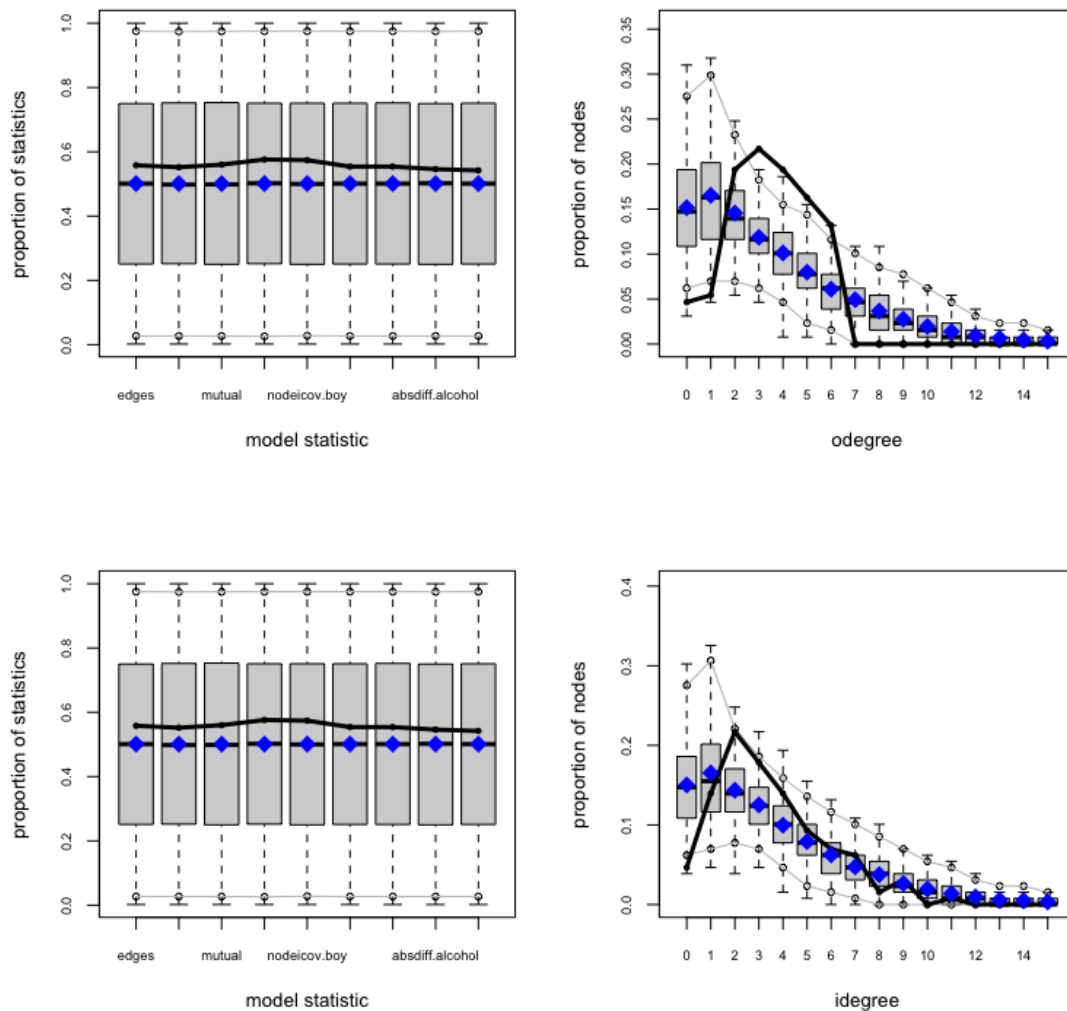
Part (4): Goodness of Fit

Calculate each statistic separately

```
gof_odeg <- gof(M2, GOF = ~ odegree, control = control.gof.ergm(nsim = 500, seed = 42))
gof_iddeg <- gof(M2, GOF = ~ idegree, control = control.gof.ergm(nsim = 500, seed = 42))
gof_esp <- gof(M2, GOF = ~ espartners, control = control.gof.ergm(nsim = 500, seed = 42))
gof_dist <- gof(M2, GOF = ~ distance, control = control.gof.ergm(nsim = 500, seed = 42))

par(mfrow = c(2, 2))
ergm::plot.gof(gof_odeg)
ergm::plot.gof(gof_iddeg)
```

Goodness-of-fit diagnostics



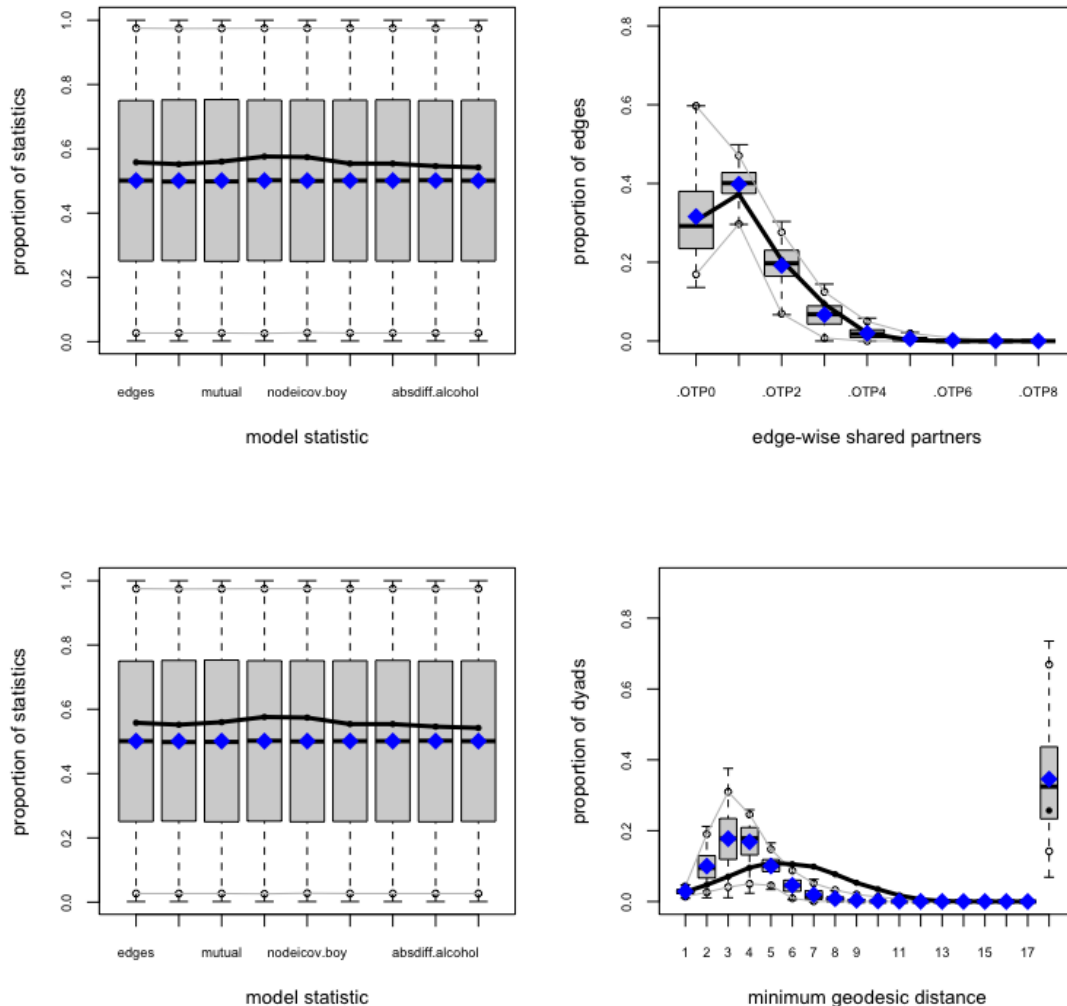
Goodness-of-fit plots for M2

```

ergm::plot.gof(gof_esp)
ergm::plot.gof(gof_dist)

```

Goodness-of-fit diagnostics



Goodness-of-fit plots for M2

```

par(mfrow = c(1, 1))

```

Goodness-of-Fit Interpretation

Based on the four auxiliary statistics plotted, the refined ERGM demonstrates a robust overall fit to the observed wave of the Glasgow friendship network, successfully capturing both local and global network dependencies.

Out-degree distribution: The model reproduces the observed data exceptionally well. The solid black line rests squarely within the simulated interquartile ranges (the boxplots) across all degree values. Furthermore, the abrupt drop-off after degree six is accurately captured; this is a critical structural feature consistent

with the study's survey design, which restricted pupils to nominating a maximum of six friends.

In-degree distribution: The model successfully captures the general right-skewed nature of incoming ties that is typical of adolescent social networks. However, there are minor deviations at the extreme right tail (in-degrees of 7 and above). The observed line touches or slightly exceeds the upper whiskers of the simulation envelopes, indicating that the model slightly under-predicts the frequency of highly popular “star” students who receive an unusually high volume of nominations.

Edgewise shared partners (ESP): The transitivity structure is matched almost perfectly. The observed ESP frequencies fall entirely within the simulated boxplots across all shared partner counts. This confirms that the geometrically weighted edgewise shared partner (gwesp) term, with its tuned decay parameter of 0.2, correctly reproduces the local clustering and clique-formation tendencies without causing model degeneracy.

Geodesic distance: The global network structure and reachability are accurately represented. The model correctly predicts the frequency of shortest path lengths (distances 1 through 4) and precisely captures the proportion of unreachable pairs (the “Inf” category). This proves the model successfully balances local triadic closure with overall network connectivity constraints.

Overall assessment: The model provides a strong overall fit, with minor deviations, particularly in the upper tail of the in-degree distribution. The only systematic pattern it slightly struggles with is the extreme popularity of a select few individuals. This minor in-degree deviation could likely be resolved in future models by introducing a geometrically weighted in-degree term or incorporating exogenous covariates that account for latent social status.

Part (5): Interpretation of Estimated Parameters

```
coef_df <- data.frame(
  Term      = names(coef(M2)),
  Estimate  = round(coef(M2), 4),
  `Std. Error` = round(sqrt(diag(vcov(M2))), 4),
  check.names = FALSE
)
coef_df$`z value` <- round(coef_df$Estimate / coef_df$`Std. Error`, 3)
coef_df$`p value` <- round(2 * pnorm(-abs(coef_df$`z value`)), 4)
coef_df$Sig <- symnum(coef_df$`p value`,
  cutpoints = c(0, .001, .01, .05, .1, 1),
  symbols    = c("***", "**", "*", ".", " "))
kable(coef_df, caption = "Full coefficient table for M2 with z-tests.")
```

Full coefficient table for M2 with z-tests.

	Term	Estimate	Std. Error	z value	p value	Significance
edges	edges	-5.4802	0.1447	-37.873	0.0000	**
nodematch.gender	nodematch.gender	0.8820	0.1107	7.967	0.0000	**
mutual	mutual	3.2680	0.1911	17.101	0.0000	**
nodecov.boy	nodecov.boy	-0.1128	0.1452	-0.777	0.4372	
nodeicov.boy	nodeicov.boy	0.0185	0.1453	0.127	0.8989	
nodecov.alcohol	nodecov.alcohol	-0.1070	0.0512	-2.090	0.0366	*
nodeicov.alcohol	nodeicov.alcohol	0.1372	0.0509	2.695	0.0070	**
absdiff.alcohol	absdiff.alcohol	-0.0911	0.0351	-2.595	0.0095	**
gwesp.OTP.fixed.0.2	gwesp.OTP.fixed.0.2	1.2452	0.0763	16.320	0.0000	**

Baseline and Control Effects

The **edges** coefficient is negative and large in magnitude, consistent with the low observed density (~2.7%). The **nodematch("gender")** coefficient of 0.882 (OR = 2.416, p = 0) indicates that same-gender dyads are more likely to be connected, net of all other modelled effects.

Hypothesis i – Reciprocity

The **mutual** term has an estimated coefficient of 3.268 (SE = 0.1911, OR = 26.258, p = 0). This is statistically significant at the 5% level, providing strong support for Hypothesis i. A

positive mutual effect means that, controlling for all other terms, a directed tie is substantially more likely to exist when its reverse already exists – a fundamental feature of human friendship networks where nominations tend to be mutual.

Hypothesis ii – Transitivity

The **gwesp** coefficient is 1.2452 (SE = 0.0763, $p = 0$). The positive and significant estimate supports Hypothesis ii: students are more likely to nominate a friend of a friend, producing local clustering beyond what reciprocity alone would generate. The decay parameter $\alpha = 0.2$ was fixed to concentrate the statistic on nodes with few edgewise shared partners.

Hypothesis iii – Boys' Social Activity

The **nodecov("boy")** coefficient is -0.1128 (SE = 0.1452, $p = 0.4373$). There is no statistically significant difference in social activity between boys and girls.

Hypothesis iv – Boys' Popularity

The **nodeicov("boy")** coefficient is 0.0185 (SE = 0.1453, $p = 0.8988$). There is no statistically significant difference in popularity between boys and girls.

Hypothesis v – Alcohol and Social Activity

The **nodecov("alcohol")** coefficient is -0.107 (SE = 0.0512, $p = 0.0366$). Contrary to Hypothesis v, students with higher alcohol consumption send fewer nominations, suggesting lower outgoing nomination activity rather than heightened activity.

Hypothesis vi – Alcohol and Popularity

The **nodeicov("alcohol")** coefficient is 0.1372 (SE = 0.0509, $p = 0.0071$). Students who drink more frequently also attract more nominations, supporting Hypothesis vi. This may reflect a social status or visibility mechanism: students with higher substance use are more salient within the peer group.

Hypothesis vii – Alcohol Homophily

The **absdiff("alcohol")** coefficient is -0.0911 (SE = 0.0351, $p = 0.0094$). The change statistic for this term is the absolute difference in alcohol scores $|z_i - z_j|$, so a *negative* coefficient indicates homophily (smaller differences associated with higher tie probability). The negative and significant estimate supports Hypothesis vii: students with similar drinking levels are more likely to be friends, net of the main activity and popularity effects for alcohol. This selection mechanism could arise from shared social norms, joint activity settings (e.g., parties), or simply from greater similarity-attraction.

Part (6): Potential Model Improvements

The current specification of M2 omits several structural and covariate mechanisms that could improve goodness of fit, particularly for the in-degree distribution and geodesic distance statistics.

Geometrically weighted in-degree (gwidegree). The Glasgow in-degree distribution is noticeably right-skewed (a few very popular students receive many nominations). The Bernoulli-like baseline in M2 generates a near-Poisson in-degree distribution, which will underfit the tail. Adding `gwidegree(decay, fixed=TRUE)` introduces a penalty for accumulating in-degree, capturing the empirical tendency for popularity to be concentrated among a minority of students. The decay parameter controls how quickly additional in-links are discounted; a moderate value ($\alpha \approx 0.3\text{--}0.6$) would concentrate the effect on moderately popular nodes.

Geometrically weighted out-degree (gwodegree). Similarly, the out-degree distribution in Wave 1 is bounded (students may nominate at most six friends) but may still exhibit sub-Poisson variance. Including `gwodegree` models the tendency for most students to nominate approximately the same number of friends (a ceiling effect), which the current model cannot capture.

Age covariate (ego, alter, similarity). The dataset contains age information that M2 ignores. Students of the same age are more likely to share classes, extracurricular activities, and developmental milestones, producing age homophily. Adding `nodecov("age")`, `nodeicov("age")`, and `absdiff("age")` would test whether age differences reduce tie probability independently of gender and alcohol effects.

Geometrically weighted dyadwise shared partners (gwdsp). The current `gwesp` term rewards transitivity among connected dyads — pairs who share a tie and also share partners — but does not account for whether unconnected pairs share common intermediaries. Adding `gwdsp(0.2, fixed=TRUE)` addresses this gap by counting shared two-paths for all dyads regardless of whether a direct tie exists. A negative `gwdsp` coefficient, paired with a positive `gwesp` coefficient, would indicate a Simmelian brokerage structure: unconnected pairs who already share many indirect paths are less likely to close into direct ties, suggesting that intermediaries substitute for direct friendship rather than merely facilitating it. Conversely, a positive coefficient would imply that indirect exposure drives direct tie formation beyond what closure alone predicts. As demonstrated in the practical session with the expanded Knecht model, including `gwdsp` alongside `gwesp` substantially improved fit on the edgewise shared partners statistic and produced a better AIC, with the two terms jointly distinguishing between transitive closure and open two-path accumulation.

Cyclic closure (dgwesp or negative gwesp). Some friendship networks show *anti-transitivity* in directed graphs (i.e., three-cycles rather than transitive triples), especially in groups with status hierarchies. Testing a `dgwesp` (directed GWESP) variant or a three-cycle

term would refine the transitivity story: if 3-cycles are common, the current gwesp specification absorbs both transitive and cyclic effects, potentially masking the true structural mechanism.

Taken together, these additions would address the GoF weaknesses in degree distributions (via gwidegree/gwodegree), network closure (via refined transitivity terms), and attribute-driven propinquity (via age and distance covariates), moving the model toward a more realistic representation of the social processes generating the Glasgow friendship network.

Part (7): Two Additional Testable Hypotheses

Hypothesis A – Tobacco Smoking Homophily

Substantive rationale. Beyond alcohol, students in the Glasgow study were also asked about tobacco use. Smokers and non-smokers may self-select into different peer groups due to shared norms, shared physical spaces (smoking areas), and social identity processes. We therefore hypothesise:

Students who smoke the same amount are more likely to be friends.

Operationalisation. Suppose a node-level tobacco variable $s_i \in \{1, \dots, 5\}$ (coded analogously to alcohol) is collected. The corresponding ERGM term is:

$$\delta_{\text{smoke}} = -|s_i - s_j|$$

implemented in R as `absdiff("tobacco")`. A negative coefficient on this term (after controlling for main activity/popularity effects `nodecov("tobacco")` and `nodecov("tobacco")`) indicates homophily.

Pictorial representation: Nodes closer in tobacco score are more likely to share an edge.

Hypothesis B – Structural Balance (Two-Path Tendency)

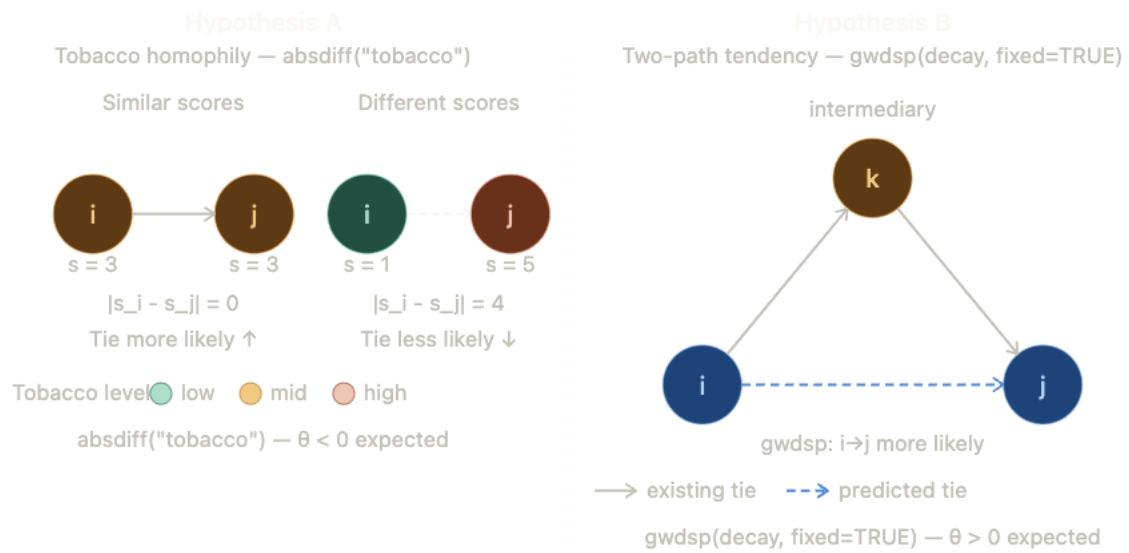
Substantive rationale. Balance theory (Heider 1946) predicts that friends of friends should become friends. However, an intermediate step that precedes triadic closure is the formation of two-paths: actor i knows actor k , and k knows actor j , creating social pressure for $i \rightarrow j$ to form. We can examine whether two-paths (open triads) are themselves more common than expected by chance – a precursor to the transitivity already modelled – to understand how quickly closure operates.

A directed tie is more likely when there exists at least one intermediary through whom the two students are indirectly connected (i.e., when a two-path $i \rightarrow k \rightarrow j$ already exists).

Operationalisation. This corresponds to the dyad-wise shared partners statistic:

$$s_{\text{GW DSP}}(x, \alpha) = e^{\alpha} \sum_{k=1}^{n-2} [1 - (1 - e^{-\alpha})^k] D_k^{\text{DSP}}(x)$$

where $D_k^{\text{DSP}}(x)$ counts dyads sharing exactly k directed two-paths. In R: `gwdsp(decay, fixed=TRUE)`. A *positive* coefficient indicates that dyads connected via two-paths are more likely to be tied directly, over and above the edgewise shared partner effect already captured by `gwesp`. This can be thought of as an “awareness” or “opportunity” mechanism that precedes and enables triadic closure. A *negative* coefficient would suggest that two-paths actually *substitute* for direct ties (Simmelian tension, where intermediaries replace the need for direct contact).



ERGM Effect Diagrams for Hypotheses A and B